



EMG-based speech recognition using dimensionality reduction methods

Anat Ratnovsky¹ · Sarit Malayev^{1,2} · Shahar Ratnovsky^{3,4} · Sara Naftali¹ · Neta Rabin⁵

Received: 1 July 2020 / Accepted: 15 May 2021 / Published online: 23 May 2021

© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2021

Abstract

Automatic speech recognition is the main form of man–machine communication. Recently, several studies have shown the ability to automatically recognize speech based on electromyography (EMG) signals of the facial muscles using machine learning methods. The objective of this study was to utilize machine learning methods for automatic identification of speech based on EMG signals. EMG signals from three facial muscles were measured from four healthy female subjects while pronouncing seven different words 50 times. Short time Fourier transform features were extracted from the EMG data. Principle component analysis (PCA) and locally linear embedding (LLE) methods were applied and compared for reducing the dimensions of the EMG data. K-nearest-neighbors was used to examine the ability to identify different word sets of a subject based on his own dataset, and to identify words of one subject based on another subject's dataset, utilizing an affine transformation for aligning between the reduced feature spaces of two subjects. The PCA and LLE achieved average recognizing rate of 81% for five words-sets in the single-subject approach. The best average recognition success rates for three and five words-sets were 88.8% and 74.6%, respectively, for the multi-subject classification approach. Both the PCA and LLE achieved satisfactory classification rates for both the single-subject and multi-subject approaches. The multi-subject classification approach enables robust classification of words recorded from a new subject based on another subject's dataset and thus can be applicable for people who have lost their ability to speak.

Keywords Electromyography · Speech recognition · Automatic speech recognition · Machine learning algorithms · Feature extraction · Principal component analysis

1 Introduction

The most common form of man–machine interaction is automatic speech recognition (ASR) which is based on acoustic signals. In ASR, the acoustic speech is decoded into a sequence of tokens, such as words, using classification techniques (Meltzner et al. 2008). The development of ASR has led to many applications such as speech forensics, robotics, and home control systems (Dhakal et al. 2019), however, ASR also has some limitations. First, the performance of ASR decreases in the presence of acoustic noise, which makes this method unsuitable for a noisy environment. Second, for word identification applications, which are used for example in the military and security domain to detect specific commands, using ASR may expose sensitive data. Lastly, since acoustic speech is audible, people who have lost their ability to speak normally can't use ASR interfaces (Meltzner et al. 2008; Srisuwan

✉ Neta Rabin
netara@tauex.tau.ac.il

Anat Ratnovsky
ratnovskya@afeka.ac.il

Sara Naftali
saran@afeka.ac.il

¹ School of Medical Engineering, Afeka-Tel Aviv Academic College of Engineering, Tel Aviv, Israel

² School of Neuroscience, Tel-Aviv University, Tel Aviv, Israel

³ School of Electrical Engineering, Tel Aviv University, Tel Aviv, Israel

⁴ School of Computer Science, Tel Aviv University, Tel Aviv, Israel

⁵ Department of Industrial Engineering, Tel-Aviv University, Tel Aviv, Israel

et al. 2018). This led to the need to find alternative forms of speech acquisition and recognition techniques.

Different types of sensors may be considered for acquiring speech: articulography sensors that capture the articulatory movement of the face, ultrasound and optical imaging of the vocal cords track, and electromagnetic or vibration sensors that track the glottal activity. Another alternative to speech acquisition via acoustic signals is to use signals recorded from electro-encephalographic sensors and electromyography (EMG), which measure the activity of the facial muscles (Denby et al. 2010). Since speech is produced by the activity of the facial muscles, it can be recognized from surface EMG (sEMG) signals measured from these muscles even during silent speech, overcoming the above limitations of acoustic-based ASR.

The field of EMG speech recognition gained momentum after it evolved in the mid-1980s in two studies that measured sEMG signals. The first focused on automatic classification of pronunciation of Japanese vowels (Sugie and Tsunoda 1985). The second tested the ability to automatically classify between 2, 12 and 17 given words. Discriminating between the two words resulted in a 97% correct classification rate, which dropped to 35% for 17 words (Morse and O'Brien 1986). More promising results were obtained by Chan et al. (2001) who achieved a 93% recognition rate between a ten-word vocabulary using sEMG signals from five face and neck muscles of two English speaking subjects. Manabe and Zhang (2004) achieved an average recognition rate of 64% with ten different speakers using a vocabulary of ten Japanese digits based on sEMG signals from three facial muscles. Meltzner et al. (2008) reported an average recognition rate of 92.1% using a 65-word vocabulary based on sEMG signals measured from 11 locations on the face and neck of nine American English speakers. In order to increase the communication robustness and to reduce information overload in space tasks, NASA conducted sub-vocal speech experiments. The first experiment was based on six words (forward, backward, left, right, go, stop) that were pronounced subvocally by three subjects 50 times. Several feature extraction techniques and data transformations were proposed for processing sEMG signals, for example, the short time Fourier transform (STFT) and different discrete and continuous wavelet power spectrum segmentations (Jorgensen et al. 2003; Jorgensen and Binsted 2005). An average recognition rate of 92% for individual word recognition was achieved by using a scaled conjugate gradient neural network classifier that was applied on wavelet-based features. In a continuous study, sEMG signals from two neck muscles were measured while pronouncing 15 isolated words 150 times each. The recordings were taken from a single subject who wore a standard-issue firefighting turnout jacket, a fire-retardant hood, and a Survivair Panther

SCBA unit (Betts et al. 2006). An overall average correct classification rate for the 15 words was 74%.

A continuous EMG speech recognition model was first introduced by Jou et al. (2006) for classification of sEMG signals from six facial and neck muscles measured from one subject while reading a set of 38 phonetic sentences ten times. Word error rates (WER) of 86.8% were obtained when plain spectral features, similar to those computed in regular ASR, were used. The use of EMG-based features significantly reduced the WER. Several studies on continuous speech recognition were done using sEMG signals measured from five neck and facial muscles. The studies were done aiming to compare speaker dependent, speaker independent and speaker adaptive models (Wand and Schultz 2009), to examine the session independent and session adaptive system (Wand and Schultz 2011) and to investigate the use of a supervised model vs. an unsupervised one in session independent recognizer (Wand and Schultz 2014).

More recent studies examined the discrepancy between different speaking modes in EMG based speech recognition. Wand et al. (2014) used sEMG signals measured from 78 subjects during two sessions, one audible and one silent. In each session each subject spoke 40 different sentences and 10 base (repeated) sentences. An average WER of 21% and 36% were obtained for the audible and silent mode, respectively, when the system was trained based on audible and silent EMG using time domain and bundled phonetic features (BDPF). Recently, a speech-related algorithm was designed based on features extracted from sEMG signals of ten neck and facial muscles to recognize isolated words, a vocabulary of continuous phrases of small words, and a large vocabulary of unseen words based on phoneme models, all pronounced vocally and silently (Meltzner et al. 2018). In another study, the authors examined the best features from among 22 that were extracted from sEMG signals of seven neck and facial muscles. An average of 93% and 80% accuracy rates for recognition of 11 Thai digits were reached with audible and silent speech, respectively (Srisuwan et al. 2018).

While all the above previous studies have shown the potential of speech recognition based on sEMG signals of several neck and facial muscles of isolated words, during continuous speech and even during whispered or silent speech, the issue of recognizing isolated words pronounced by one subject based on a dataset constructed from signals of another subject still needs to be resolved.

The goal of this work is twofold: first to analyze the ability to classify isolated words based on sEMG from only three facial muscles for a single subject. Within this analysis, we compare between the performances of a nonlinear dimensionality technique, which are able to capture the intrinsic structure of the data (Liu et al. 2016; Rabin et al. 2019), to a standard linear dimensionality method. In particular, we

examine the classification rates of a principal components analysis (PCA) and a locally linear embedding (LLE) based algorithm for different combinations of words.

The second goal of this work is focused on classification of data recorded from one new (test) subject based on training signals that are recorded from another subject. This point gained less attention in previous research and has rarely been studied or addressed in most of the field of speech recognition based on sEMG signals. It is in principal a more challenging problem, as each person introduces different variability to the dataset. One way to overcome the variability between subjects is to gather a very large training set. Here, a different approach, which relays on data alignment, is proposed. In particular, we align a small subset of labeled data recorded from one test subject to a space that holds many labeled training signals that were recorded from another (train) subject. This algorithm aims to bypass the variance between subjects, without collection of massive training sets.

1.1 Mathematical background

In this section a brief essential mathematical background for the dimensionality reduction and data alignment methods that were utilized in this study, is provided.

1.1.1 Dimension reduction methods

PCA is a standard method for dimensionality reduction. In PCA, the dimension of a given dataset is reduced while retaining most of the original linear variability in the data. In practice, the covariance matrix of the data is computed and the dataset is projected onto the top eigenvectors of the matrix (Pearson 1901; Jolliffe 1986).

LLE is a nonlinear dimensionality reduction technique that computes a low-dimensional embedding of a given high-dimensional dataset while preserving the local structures of the original data (Roweis and Saul 2000). Let $X = \{x_1, \dots, x_N\}$ be a set of high-dimensional data points, such that $x_i \in \mathbb{R}^D$. In order to faithfully describe the local geometry of the dataset, the closest K neighbors of each data point x_i are computed. Then, the nearest neighbors approximate each data point x_i by computing a linear combination that minimizes the reconstruction errors, as follows

$$\varepsilon(W) = \sum_{i=1}^N \left[x_i - \sum_j W_{ij} x_j \right]^2. \quad (1)$$

The entries $W_{ij} = w(x_i, x_j)$ are computed only for points x_j that were identified as neighbors of x_i , otherwise, $w(x_i, x_j) = 0$. In addition, the weight vector is normalized such that $\sum_j w(x_i, x_j) = 1$. The weights are determined by computing a least square problem. The $d \ll D$

low-dimensional coordinates, which are denoted by $\Phi = \{\phi_1, \dots, \phi_d\}$ where $\Phi_i = \Phi(x_i) = (\phi_1(x_i), \dots, \phi_d(x_i))$, are those which minimize the quadratic form in the following Eq. (2)

$$C(\Phi) = \sum_{i=1}^N \left[\Phi_i - \sum_j W_{ij} \Phi_j \right]^2. \quad (2)$$

Since the same weights are used in both Eqs. (1) and (2), the new embedding coordinates serve as a compact neighborhood preserving map for the high-dimensional data.

1.1.2 Affine transformation method

An affine transformation algorithm (Lafon et al. 2006) aligns two embedded datasets based on a subset of matching points that are referred to as landmarks (Rabin et al. 2020). The affine transformation is calculated in the d -dimensional embedding space. It is also possible to calculate the transformation based on a larger number of embedding coordinates for gaining more degrees of freedom; then the calculated rotation is projected onto the desired number of coordinates, d . In detail, let $Y = \{y_1, y_2, \dots, y_{N_1}\}$ and $Z = \{z_1, z_2, \dots, z_{N_2}\}$ such that $\{y_i\}_{i=1}^{N_1} \in \mathbb{R}^D$ and $\{z_i\}_{i=1}^{N_2} \in \mathbb{R}^D$ are the results of the PCA or LLE dimensionality reduction application to a d -dimensional space. The alignment is based on L landmark pairs of points in Y and Z , denote these by $\{(y_1, z_1), (y_2, z_2), \dots, (y_L, z_L)\}$. An affine function $g : Y \rightarrow Z$ that has the form $g(y_i) = Mz_i + b$, where M is a linear transformation and b is a vector, defines the transformation. Moreover, for the landmark points it is requested that g satisfies $g(y_1) = z_1, g(y_2) = z_2, \dots, g(y_L) = z_L$. Finally, the transformation function is derived by computing $\arg \min_{z \in Z} \{ \|g(y_i) - z\| \}, \quad y_i \in Y$.

1.2 Experimental apparatus and protocol

The experimental apparatus included three EMG amplifiers (BIPAC EMG100A), disposable AG–AGCL surface electrodes, an A/D board (PCI-6211) and a PC equipped with LabVIEW software for windows. EMG signals from the orbicularis oris, depressor anguli oris and mylohyoid, which are three facial muscles, were measured noninvasively using surface EMG electrodes from four female subjects (non-native English speakers). The experiments, who were performed by a biomedical graduate student, were carried out as follows: two electrodes were placed under the chin, in the upper part of the neck, for measuring the submental activity over the mylohyoid (Ding et al. 2002; Lee et al. 2015); two electrodes were placed near the lips, for measuring the activity of the orbicularis oris (Lapatki et al. 2003; Ratnovsky et al. 2013); two electrodes were placed midway between the modiolus and the mandible (Lapatki et al. 2003)

and one electrode was placed on the back of the hand for reference (Fig. 1).

Typically, the frequency contents of raw surface EMG can range between 6 and 500 Hz, while most of the frequency power lies between ~ 20 and 150 Hz (Konrad 2005; Phinyomark and Scheme 2018). Thus, a sampling rate of 1000 Hz was used to correctly “decode” the frequency spectrum of the EMG signal. Other EMG based speech recognition studies used similar and even lower sampling rate (Jong and Phukpattaranont 2019; Srisuwan et al. 2018; Wand and Schmidhuber 2016).

The EMG signals were measured while the subject was sitting and pronouncing the following 11 isolated words: forward, backward, left, right, go, stop, one, two, three, four and five, 50 times each. Each subject performed eleven sessions, one for each word. At each session the same isolated word was repeated for 50 times. The subjects rested for at least 10 min between successive sessions, the electrodes were not removed or replaced between sessions of the same subject. We note that during these experiments and analysis, only the EMG signals were recorded and we did not rely on the voice signals. Therefore, until the end of this paper, the meaning of *word* relates to its EMG signal realization, regardless of the voice that may or may not have been pronounced by the subjects.

1.3 Data analysis

1.3.1 EMG segmentation

Each raw EMG signal included all the repetitions taken from each word. To yield only the muscle activity, each raw EMG signal was manually segmented according to the first activated muscle at the time of word pronunciation. (Fig. 2). The vertical solid lines schematically represent

the areas where the full signal is divided into segments. This led to 50 separate signals per muscle for each word. In total, 150 EMG signals per word were generated for each subject. Different segments represent different repetitions and as a result have a different length. Therefore, each signal was padded with zeros according to the length of the longest signal from all possible ones associated with the 11 words. In average, the majority of the signals had the same length. Nevertheless, a small number of signals that were substantially shorter than the longest signal and required significant padding, were removed from the data-set. After padding, all signals had the same vector length of 3970.

1.3.2 Features extraction

Each EMG signal was transformed to the frequency domain by computing the short time Fourier transform (STFT). Specifically, after examination of three possible window sizes: 64, 128, 256 and two window functions: Hamming and Hann, the chosen window was Hann with window size of 128 and an overlap of 64. This yields a feature matrix of size 33×126 , which describes the frequency distribution of the signal as a function of time. The STFT based feature matrix was then row-stacked to one long vector of 4158 features for each segment. Each vector holds the features of one EMG signal from one muscle. For each word repetition, three EMG signals were recorded, one from each recorded facial muscle. Therefore, three corresponding features vectors were calculated for each word repetition and then concatenated to form one united feature matrix (4158×3) of 12,474 features. This led to 50 STFT vectors for each word per subject.

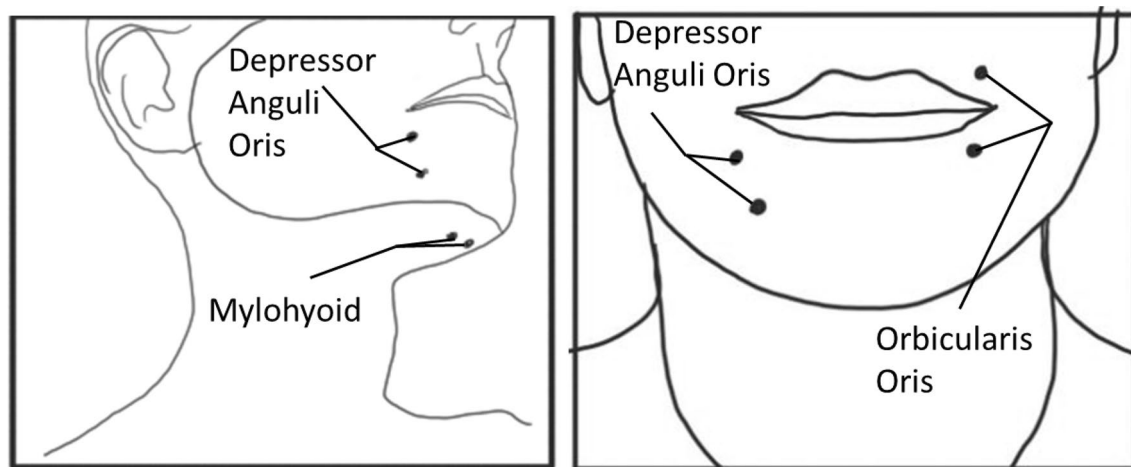
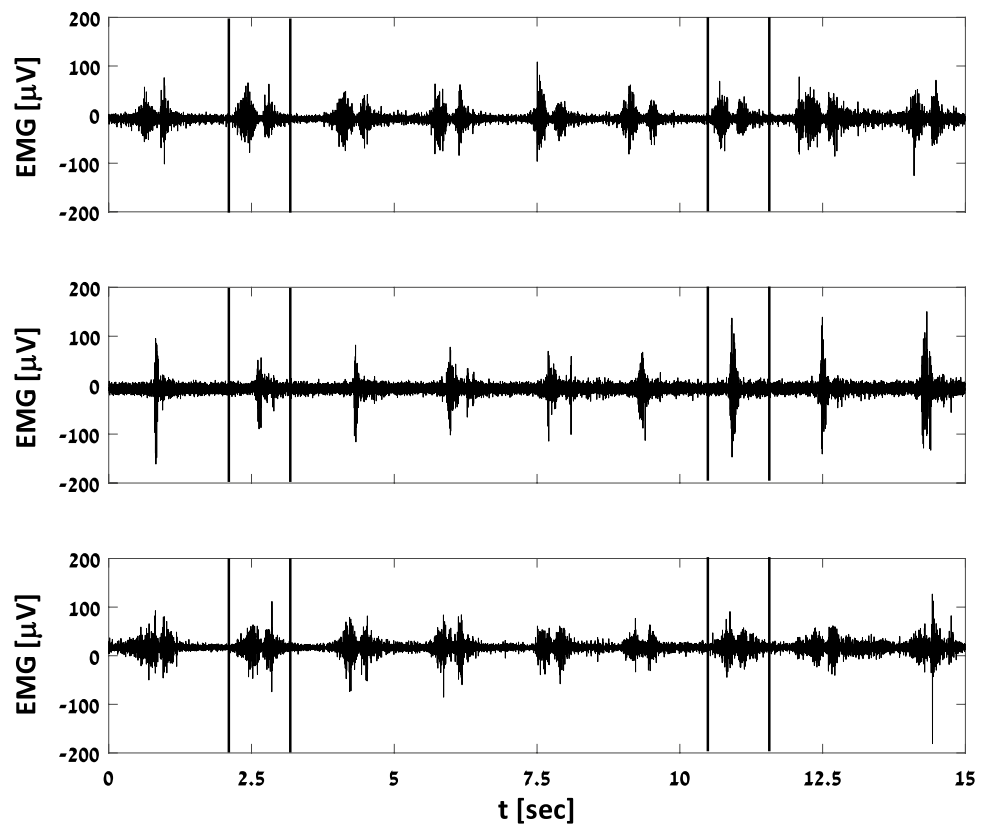


Fig. 1 EMG electrodes position

Fig. 2 An example of the raw EMG signals. The vertical solid lines schematically represent the areas where the full signal is divided into segments. Segmentation was done according to the first activated muscle at the time of word pronunciation



1.3.3 Dimensionality reduction

The features matrix constructed for each subject included 12,474 features for each word-sample. In order to generate a compact representation, a dimensionality reduction algorithm was applied on the datasets. PCA and LLE methods were used in order to reduce the dataset.

In this work, the original high-dimensional set is the feature space that is computed for each word. The lower dimension, d , for both LLE and PCA was set to 15, after testing several values for d (5, 10, and 15) in all of the experimental settings. It should be noted that one hand it is desirable to reduce the dimension and end up with a very small number of coordinated, however, since later in this paper the an affine transformation is computed based on the reduced coordinated, a very small number of coordinates degrade the results of this step. The new features are derived by linear (for PCA) or non-linear (for LLE) combinations of the STFT frequency values. Figure 3a and b show an example of three-dimensional (3D) scatter of the STFT low-dimensional features set for five words (one, left, three, right, go) after PCA and LLE methods were applied. Each color represents a specific word and each point represents a sample in the features data set.

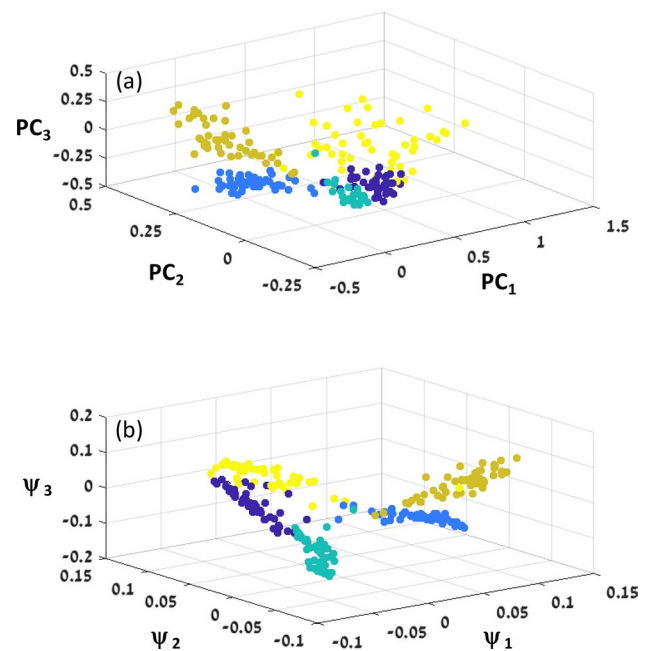


Fig. 3 A 3D scatter of **a** the STFT features set after PCA, **b** the STFT features set after LLE. Each color represents a different word

1.4 Classification of words for each subject based on its own dataset

The ability to identify unlabeled words of one subject based on its own dataset was first investigated. The classification was based on datasets of different combinations of five words. Both the training and the test datasets were composed of word-samples from the features matrixes. In all cases, the training dataset included 90% of the data and the test dataset included 10% of the data. In every classification fold, five word-samples from each word set were randomly chosen for the test dataset and the remaining 45 word-samples from each word were a part of the training dataset. For each subject, two methods, PCA and LLE were applied and compared for reducing the dimensions of the feature set followed by a K-nearest-neighbors (KNN) classification. Each algorithm was applied ten times on three different word-sets: word-set #1: go, stop, left, forward, backward, word-set #2: go, one, two, three, four and word-set #3: one, backward, left, three, five. An average recognition rate was calculated.

1.5 Classification of words for each subject based on another subject's dataset

There exists a large variance between the datasets of different subjects due to the fact that each subject has his own unique muscle variability and since the location of the EMG electrodes for each muscle between different subjects cannot be exactly the same. Therefore, in order to classify words pronounced by one subject based on a dataset of another subject, one tested dataset must be aligned according to the coordinates of the other dataset. An affine transformation algorithm (Lafon et al. 2006) was used in order to project the test dataset of one subject to the training dataset of another subject. In this application, the affine transformation is computed during the training phase, and then applied in the classification phase for new test words. The STFT features, which are computed for the training set are input for the dimensionality reduction algorithm. Then, the alignment is computed in the low-dimensional space, which is the result of the PCA or LLE algorithms.

The alignment algorithm was applied to rotate and merge the low-dimensional representation of the test subjects #1, #2 and #4 onto the low-dimensional representation of the training subject #3 and then to rotate and merge the low-dimensional representation of the test subject #3 onto the low-dimensional representation of the training subject #1.

Classification, for example, of five words that belong to subject #1 given the full labeled dataset of subject #3, was done in the following manner:

Subject #1 has 20 labeled word-samples of each of the five words, which resulted in a total of 100 labeled feature vectors. The training set subject holds 250 labeled feature

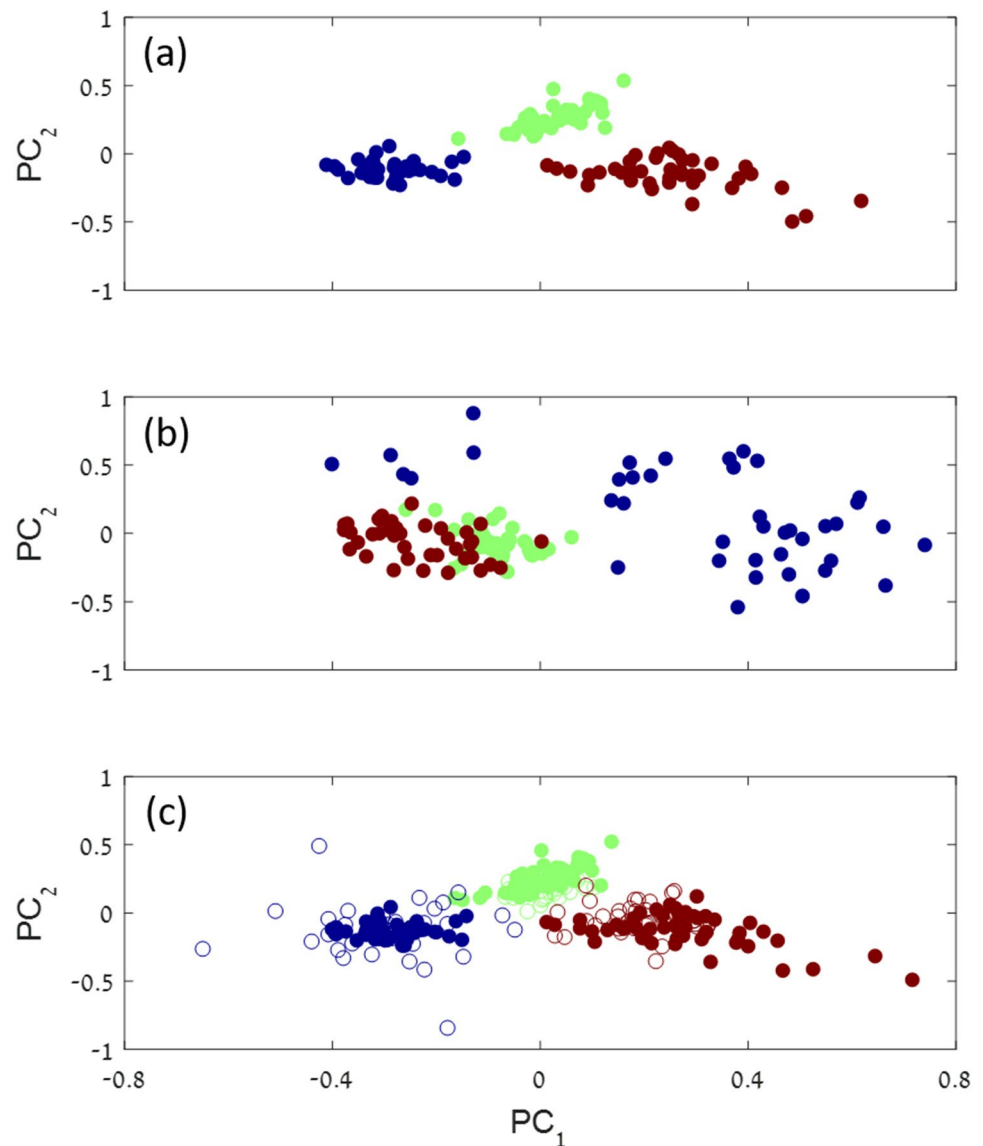
vectors (50 for each word). Each subject's feature space was independently embedded into a low-dimensional space. Then, the embedding of test subject #1 was aligned to the embedding by randomly choosing 20 landmark points from the training subject's embedding. The rest of the test samples were aligned according to the calculated transformation matrix. An example of the low-dimensional representations belonging to the training and test subjects before and after the alignment of the test subject on the training subject is seen in Fig. 4. The filled points in Fig. 4c belong to the training dataset and the hollow points belong to the test dataset. Each color in the Fig. 4 represents a different word. In addition, for the alignment of the test dataset of subject #1 on the training dataset of subject #3 and test dataset of subjects #3 on the training dataset of subject #2, the influence of the transformation matrix was also investigated and the number of landmark points for the calculation of the transformation matrix coefficients was gradually decreased from 20 to 10, 7 and 5.

Finally, classification of a test word-sample (that belongs to subject #1) was computed as follows: STFT features were extracted from a new word-sample and the feature vector was embedded in the low-dimensional (PCA or LLE) space that belongs to the test subject. Then, the computed affined transformation was applied to the new embedded test point, this rotated the test point into the training subject's embeddings. KNN was then used to classify the new embedded test point. The test dataset, here, included 40 word-samples from each word and the training dataset included all 50 existing word-samples from each word. The recognition of three word-sets: forward, five, stop; five word-sets: backward, stop, left, right, three; and seven word-sets: backward, stop, left, right, one, two, three, were investigated for each subject, while the training dataset belonged to a different subject. At each classification round, a tenfold cross validation was applied, at each round the word-samples from each word were randomly chosen for the test dataset.

2 Results

EMG signals measured from three facial muscles were acquired from four female adults while pronouncing the following words: backward, forward, go, stop, right, left, one, two, three, four, and five. Pronunciation of each word was repeated 50 times, segmentation led to a total of 150 EMG signals for each word per subject. STFT features were extracted from each segment, PCA and LLE were applied on the STFT features set of each subject. An example of a 3D scatter of the STFT features set of five words after applying PCA and after applying LLE is presented in Fig. 3a and b, respectively. It can be seen from the 3D scatter of the STFT that after applying both the PCA and LLE, there is a

Fig. 4 Alignment of a 3-word test dataset of one subject to the 3-word training dataset of another. Each color represents a different word. **a** Subject #1, **b** subject #2, **c** subject #2-hollow circle, aligned on subject #1-solid circle



separation between different zones of the word-samples of each of the five words.

The ability to differentiate between a set of five words was tested for each subject using KNN classification, several 5-word sets were tested. The third word-set (one, backward, left, three, five) achieved the highest average recognition rates of 81% compared to an average value of 73% in the first and second word-sets (Fig. 5). In all the three word sets, the LLE method yields slightly higher success rates compared to PCA.

The ability to classify words of one subject based on a dataset constructed from signals of another subject was investigated on datasets of three words, five words and seven words. Each test dataset was aligned to the training dataset using an affine transform (Fig. 6). The best average recognition success rates for three words-set, five words-set and

seven words-set were 88.8%, 74.6% and 49.6%, respectively (Fig. 7).

The classification of five words was also examined during the decrement of the number of landmark points used to calculate the transformation matrix, as described in Fig. 8. The recognition success rates decreased from 82% for 20 landmark points to 61% for five landmark points.

3 Discussion

The presented research aimed to recognize and classify speech based on EMG signals from only three facial muscles. The EMG database was constructed from measurements of four female subjects. Each subject pronounced 11 words 50 times. The recorded words included basic

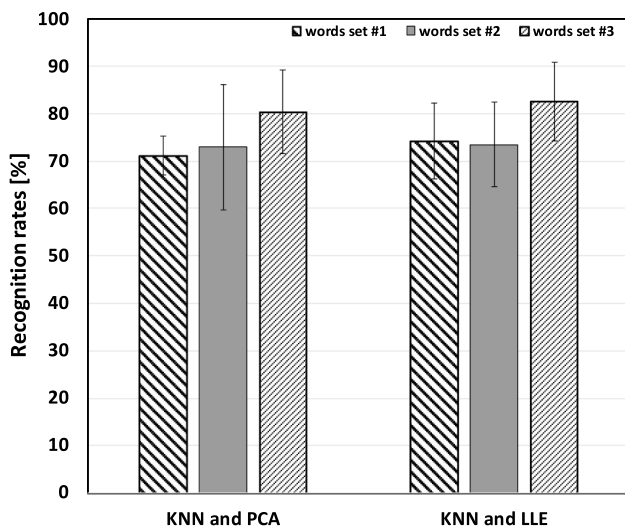


Fig. 5 Average $\pm$ SD recognition success rates of three different word-sets using two methods of dimension reduction

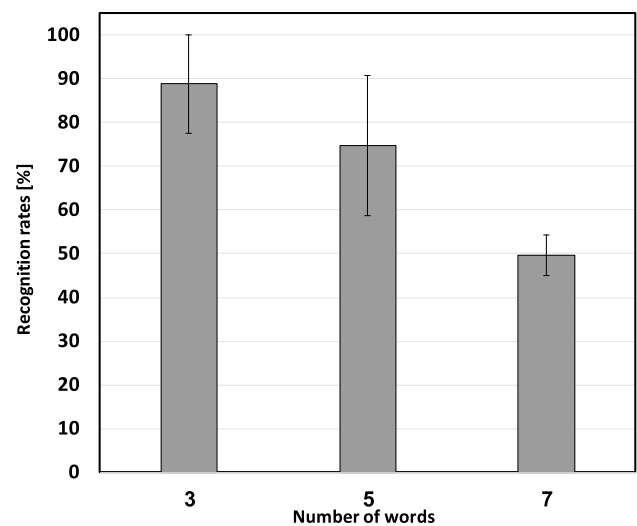


Fig. 7 Total average recognition success rates as function of the number of words

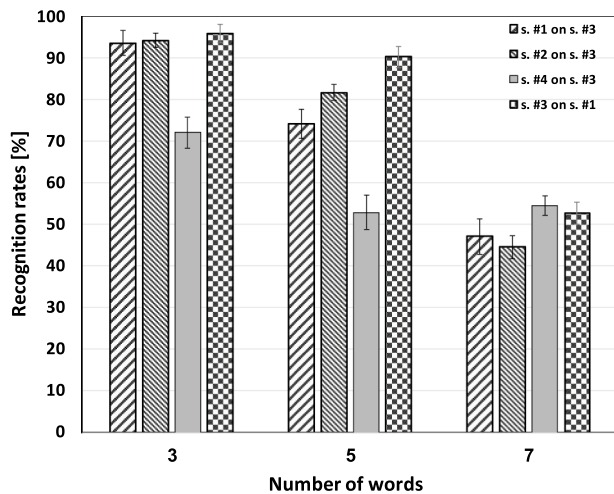


Fig. 6 Average $\pm$ SD recognition success rates for three words-set, five words-set and seven words-set using an affine transform method while aligning the low-dimensional representation of the test subjects #1, #2 and #4 onto the low-dimensional representation of the training subject #3 and the low-dimensional representation of the test subject #3 onto the low-dimensional representation of the training subject #1

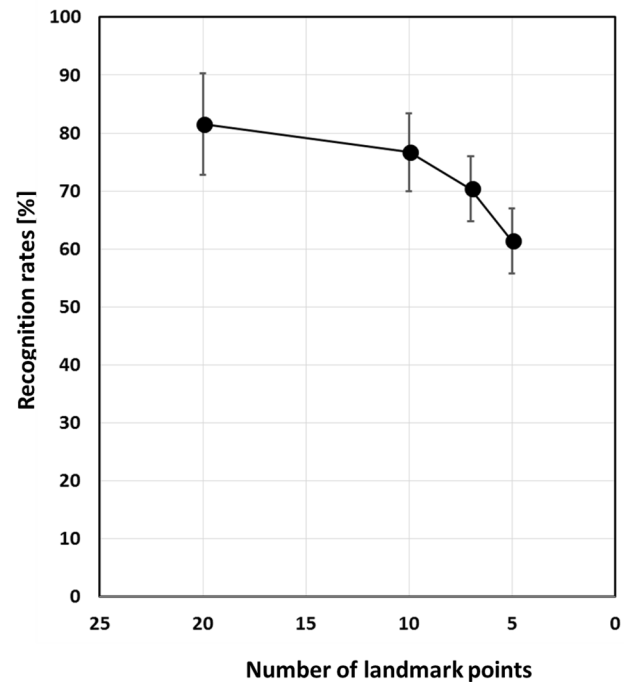


Fig. 8 Average recognition success rates of five words of two subjects as function of the number of landmark points chosen to calculate the affine transformation matrix

English words and numbers. Selection of features, such as time domain features, STFT, linear predictive codes (LPC), wavelet transform, Mel scale cepstral coefficients (MFCC), for characterizing sEMG signals for speech recognition have been the subject of several previous studies (Chan et al. 2001; Manabe and Zhang 2004; Jorgensen and Dusan 2010; Meltzner et al. 2018; Srisuwan et al. 2018). Here the STFT, which captures information both in the time and frequency domains, was selected for the feature extraction step. This feature was found previously to be significant and informative when compared to conventional EMG features

for speech recognition and other EMG application (Chan et al. 2001; Tsai et al. 2015).

PCA and LLE methods were used for dimension reduction and their performances were compared in a single subject setting. The classification was conducted on three different 5-word sets. In all the three word sets, although the LLE achieved slightly higher classification rates, non-significant

differences were obtained between the performances of the two methods.

The chosen word-set was found to have an important influence on the classification success. The third word-set (one, backward, left, three, five) showed higher recognition rates than the two other examined sets. Each of the words in this third set has a unique pronunciation, while both the first and second word-sets contain words with similar vowels. For example, in word-set #1 there is a similarity between "forward" and "backward" and in word-set #2 the similarity is between "go" and "four". Distinguishing between similar vowels is even more difficult when pronounced by non-native English speakers. Words with similar vowels require similar muscle movement and may yield similar EMG signals compared to those with completely different vowels. The average classification rate of one subject based on his own dataset obtained for the third word-set of 81% proved to be satisfactory considering that the participants were non-native English speakers. Other EMG based speech recognition studies which aimed to classify isolated words achieved lower success rates. Chan et al. (2002) achieved ~72% classification rate of a 10-word vocabulary consisting of the digits "zero" to "nine" using EMG signals of five facial muscles. Manabe and Zhang (2004), who classified ten Japanese digits pronounced by ten Japanese native speakers based also on only three facial muscles, reached 64% classification rate. Another study, which used only one EMG signal from the neck muscle to classify between 15 isolated words, achieved a classification rate of 74% based on data measured from only one subject (Betts et al. 2006). More recent studies, although reached classification rates of 92% (Jorgensen and Dusan 2010), 91% (Meltzner et al. 2018) and 93% (Srisuwan et al. 2018), used either a combination of EMG signals with lip and jaw movements (Jorgensen and Dusan 2010) or were based on EMG data measured from 7 to 11 facial and neck muscles (Meltzner et al. 2018; Srisuwan et al. 2018).

The relatively satisfactory success rates found in this study and in most of the other studies tested the setting in which both the training and the test set belonged to the same subject. However, since each subject pronounces words differently, the variance between the recorded signals is large. Data fusion for merging between several subjects is challenging and required utilization of special multi-subject models. One approach for multi-subject EMG based speech classification, as introduced by Wand and Schultz (2009), is context-independent or dependent phoneme-based recognizer. In their cross-speaker training, the system was trained on all measurements from all participating speakers except from the test speaker. It was assumed, as for the field of acoustic speech recognition, that modeling a phoneme depending on its right and left neighboring phonemes would increase the recognition accuracy since the recorded EMG signal for a given pronounced phoneme also depends on

the context in which the phoneme is spoken. Based on a data set collected from 14 speakers, the context-independent phoneme-based recognizer yields a WER of 62%, while the context-dependent yields a WER of 57%. Independent or dependent phoneme-based recognizers enable the classification of sentences and provide continuous speech classification. However, the phoneme-based recognizer requires, first, a training phase in which the EMG signal is segmented to phonemes based on the audio signal; second, the training data should be large enough to offer sufficient training samples; and third, it yields high WER for the cross-speaker training even in the context-dependent model. This high WER and the dependency on audio signals for the training phase decreases the efficiency of this approach to EMG based speech recognition for people who have lost their ability to speak normally.

In this study, a different approach of data alignment technique for multi-subject recognition was proposed using an affine transform. This approach is suited for training data of limited size and is easy to implement for constructing unified, multi-subject models. In the proposed algorithm the training subjects had a full, large training set, which included many examples of each word, while the test subjects had only a limited amount of training data. As explained in the Methods Section, the feature space of each subject was first independently embedded into a low-dimensional space and then the embedding was aligned to that of the training subject by randomly choosing landmark points from the training subject's embedding, for the calculation of the transformation matrix coefficients. The strength of the methods is clearly reflected in the classification results for three and five words-sets, where success rates of 88.8% and 74.6% were found, respectively. This robustness is also reflected in the classification results for five words-set, where 50% reduction of the landmark points from 20 to 10 points caused a reduction of only 6% in the classification results (Fig. 8). Further decrease of the landmark points reduces the classification reliability. It can be presumed that a larger group of labeled training subjects would have increased the classification success rates for any new subject. Moreover, a larger group of labeled training subjects would also enable the use of only a few landmark points from the training subjects' embedding for the calculation of the transformation matrix coefficients.

The strength of the proposed algorithm for the multi-subject classification demonstrates that this method can be applicable for people who have lost their ability to speak where it is not possible to collect a vocal training sample. For these cases, their EMG based dataset can be aligned to that measured from healthy subjects. Additionally, in the suggested method, simple implementation which requires only computing a least squares optimization, is needed.

Although the results of the present study show relatively high success rates, both when the classification is done for

each subject based on his own dataset and when the classification is done based on another subject's dataset, several issues must be considered. First, the dataset was constructed from measurements taken from only four nonnative English speaker subjects using a limited number of words. Second, the alignment algorithm was examined for only four combinations of training-test subjects. To improve the results it is recommended to increase the EMG database by measuring more subjects and increasing the number of words pronounced by each subject. In addition, the classification of words pronounced by a test subject, based on a dataset constructed from samples of more than one subject could be examined using the presented data-alignment techniques.

4 Conclusion

The results of this study clearly show that both the PCA and LLE achieved relatively good classification rates for each subject based on its own dataset. However, the third word-set achieved the highest average recognition rates and in all of the three word sets, the LLE method yields slightly higher success rates compared to the PCA. In addition, the multi-subject classification approach, which utilizes dimension reduction methods along with an alignment algorithm, enables robust classification of a new subject based on another subject's data sets even at relatively small landmark points and thus can be applicable for people who have lost their ability to speak.

References

- Betts BJ, Binsted K, Jorgensen C (2006) Small-vocabulary speech recognition using surface electromyography. *Interact Comput* 18(6):1242–1259
- Chan AD, Englehart K, Hudgins B, Lovely DF (2001) Myo-electric signals to augment speech recognition. *Med Biol Eng Comput* 39(4):500–504
- Chan AD, Englehart K, Hudgins B, Lovely DF (2002) A multi-expert speech recognition system using acoustic and myoelectric signals. In: *Proceedings of the Second Joint 24th Annual Conference and the Annual Fall Meeting of the Biomedical Engineering Society—Engineering in Medicine and Biology* 1: 72–73 IEEE
- Denby B, Schultz T, Honda K, Hueber T, Gilbert JM, Brumberg JS (2010) Silent speech interfaces. *Speech Commun* 52(4):270–287
- Dhakal P, Damacharla P, Javaid AY, Devabhaktuni V (2019) A near real-time automatic speaker recognition architecture for voice-based user interface. *Mach Learn Knowl Extr* 1(1):504–520
- Ding R, Larson CR, Logemann JA, Rademaker AW (2002) Surface electromyographic and electroglottographic studies in normal subjects under two swallow conditions: normal and during the Mendelsohn maneuver. *Dysphagia* 17(1):1–12
- Jolliffe IT (1986) *Principal components in regression analysis. Principal component analysis*. Springer, New York, pp 129–155
- Jong NS, Phukpattaranont P (2019) A speech recognition system based on electromyography for the rehabilitation of dysarthric patients: a Thai syllable study. *Biocybern Biomed Eng* 39(1):234–245
- Jorgensen C, Binsted K (2005) Web browser control using EMG based sub vocal speech recognition. In: *Proceedings of the 38th Annual Hawaii International Conference on System Sciences* 294c–294c IEEE.
- Jorgensen C, Dusan S (2010) Speech interfaces based upon surface electromyography. *Speech Commun* 52(4):354–366
- Jorgensen C, Lee DD, Agabont S (2003) Sub auditory speech recognition based on EMG signals. In: *Proceedings of the International Joint Conference on Neural Networks* 4:3128–3133 IEEE
- Jou SC, Schultz T, Walliczek M, Kraft F, Waibel A (2006) Towards continuous speech recognition using surface electromyography. In: *Ninth International Conference on Spoken Language Processing*
- Konrad P (2005) The ABC of EMG: A practical introduction to kinesiological electromyography, 30–35
- Lafon S, Keller Y, Coifman RR (2006) Data fusion and multicue data matching by diffusion maps. *IEEE Trans Pattern Anal Mach Intell* 28(11):1784–1797
- Lapatki BG, Stegeman DF, Jonas IE (2003) A surface EMG electrode for the simultaneous observation of multiple facial muscles. *J Neurosci Methods* 123(2):117–128
- Lee HY, Hong JS, Lee KC, Shin YK, Cho SR (2015) Changes in hyolaryngeal movement and swallowing function after neuromuscular electrical stimulation in patients with dysphagia. *Ann Rehabil Med* 39(2):199
- Liu Y, Zhang Y, Yu Z, Zeng M (2016) Incremental supervised locally linear embedding for machinery fault diagnosis. *Eng Appl Artif Intell* 50:60–70
- Manabe H, Zhang Z (2004) Multi-stream HMM for EMG-based speech recognition. In: *The 26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society* 2:4389–4392 IEEE
- Meltzner GS, Sroka J, Heaton JT, Gilmore LD, Colby G, Roy S, Chen N, Luca CJ (2008) Speech recognition for vocalized and subvocal modes of production using surface EMG signals from the neck and face. In: *Ninth Annual Conference of the International Speech Communication Association*
- Meltzner GS, Heaton JT, Deng Y, De Luca G, Roy SH, Kline JC (2018) Development of sEMG sensors and algorithms for silent speech recognition. *J Neural Eng* 15(4):046031
- Morse MS, O'Brien EM (1986) Research summary of a scheme to ascertain the availability of speech information in the myoelectric signals of neck and head muscles using surface electrodes. *Comput Biol Med* 16(6):399–410
- Pearson K (1901) On lines of closes fit to system of points in space, London, E dinb. *Dublin Philos Mag J Sci* 2:559–572
- Phinyomark A, Scheme E (2018) EMG pattern recognition in the era of big data and deep learning. *Big Data Cogn Comput* 2(3):21
- Rabin N, Golan M, Singer G, Kleper D (2019) Modeling and analysis of students' performance trajectories using diffusion maps and kernel two-sample tests. *Eng Appl Artif Intell* 85:492–503
- Rabin N, Kahlon M, Malayev S, Ratnovsky A (2020) Classification of human hand movements based on EMG signals using nonlinear dimensionality reduction and data fusion techniques. *Expert Syst Appl* 149:113281
- Ratnovsky A, Carmeli YN, Elad D, Zaretsky U, Dollberg S, Mandel D (2013) Analysis of facial and inspiratory muscles performance during breastfeeding. *Technol Health Care* 21(5):511–520
- Roweis ST, Saul LK (2000) Nonlinear dimensionality reduction by locally linear embedding. *Science* 290(5500):2323–2326
- Srisuwan N, Phukpattaranont P, Limsakul C (2018) Comparison of feature evaluation criteria for speech recognition based on electromyography. *Med Biol Eng Comp* 56(6):1041–1051
- Sugie N, Tsunoda K (1985) A speech prosthesis employing a speech synthesizer-vowel discrimination from perioral muscle activities and vowel production. *IEEE Trans Biomed Eng* 7:485–490

- Tsai AC, Luh JJ, Lin TT (2015) A novel STFT-ranking feature of multi-channel EMG for motion pattern recognition. *Expert Syst Appl* 42(7):3327–3341
- Wand M, Schultz T (2009) Towards speaker-adaptive speech recognition based on surface electromyography. In: *Biosignals*, pp 155–162
- Wand M, Schultz T (2011) Session-independent EMG-based Speech Recognition. In: *Biosignals* pp. 295–300.
- Wand M, Schultz T (2014) Towards real-life application of EMG-based speech recognition by using unsupervised adaptation. In: *Fifteenth Annual Conference of the International Speech Communication Association*
- Wand M, Schmidhuber J (2016) Deep neural network frontend for continuous EMG-based speech recognition. In: *Interspeech*, pp 3032–3036
- Wand M, Janke M, Schultz T (2014) Tackling speaking mode varieties in EMG-based speech recognition. *IEEE Trans Biomed Eng* 61(10):2515–2526

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.